ITAI 1371 Midterm – Individual Contribution Journal

Project: Stroke Prediction Data Preparation

For this midterm, I worked on understanding the stroke-prediction dataset, reviewing the notebook, troubleshooting problems, documenting our decisions, and organizing the project files.

I worked through the notebook in Google Colab and reviewed the different stages of the data-preparation process. This included checking the dataset, examining missing values, reviewing the stroke and non-stroke counts, and looking at graphs to understand patterns in the data. One example was learning that only about 1.8% of the records were stroke cases. This helped me understand why class imbalance is important for our project.

I also worked through our preprocessing decisions. For example, I learned why we used the median to fill missing BMI values and why we created a bmi_missing feature. I reviewed the decision to use "Unknown" for missing smoking status rather than guessing what the missing values meant. I also worked on understanding the purpose of the 70/30 train-test split and why the test data needs to stay separate from the preparation process.

Another part of my contribution was working through Python and notebook questions with AI assistance. I used explanations to help me understand code, troubleshoot errors, and make the notebook easier to follow. Rather than just copying code, I asked questions about why we were making certain decisions and how the different sections connected to the overall project.

I also worked on organizing the project through Git Bash and GitHub. I created the GitHub repository and successfully pushed the initial project files. I then worked on identifying which additional files needed to be added, including the prepared datasets and the saved preprocessing pipeline.

AI tools were part of my learning and troubleshooting process. I used them to explain technical concepts, review wording, and help me work through errors. I still needed to review the suggestions and understand how they applied to our project.

My contribution focused on learning the data-preparation process, reviewing and documenting the decisions, troubleshooting with assistance, and organizing the files for submission. This work gave me more experience using Google Colab, Python, Git Bash, and GitHub. I also have a better understanding of the work that needs to happen before we can train and evaluate the models in the final project.

Midterm Contribution Journal

ITAI 1371 Intro to Machine Learning

Group: BetaThetas

Derrick K. Gibbs-McGlaston

For the BetaThetas midterm project, I worked with the approved cerebral stroke dataset and completed the Jupyter notebook in VsCode Editor and ran it in Google Colab. My work focused on preparing the dataset and checking the results displayed by

the notebook. I also reviewed the assignment instructions to make sure the work followed the requirement to use Python rather than changing the original CSV by hand.

I ran the notebook that loaded the original 43,400 records and split them into 30,380 training records and 13,020 testing records. I checked the outputs showing that the testing data was saved separately and that the analysis was done on the training data. I examined the sample rows, missing-value counts, and graphs before cleaning.

The notebook I completed and tested filled missing BMI values with the median and marked missing smoking status values as Unknown. It also created age groups, adjusted the glucose values, converted categories into numerical columns, and scaled selected features. I used the printed samples and graphs to review how the data changed during these steps. I checked the class-balancing results because the stroke records were much less common than the non-stroke records. The training data had 29,832 non-stroke records and 548 stroke records before balancing. After the notebook repeated records from the smaller group, both classes had 29,832 rows. My own reflection focuses on what I learned about missing values, encoding, scaling, class balancing, and the need to keep testing data separate. My main contribution was preparing and checking a working version of the data-preparation notebook and documenting what I learned from the results.

Contribution Journal: Midterm Exploratory Data Analysis (EDA)
Dataset Focus: Cerebral Stroke Prediction
Team: ThetaBetas - Brandy Gonzales

Each team member reviewed the data and completed the work independently. Then we shared our files on Microsoft Teams so our team leader, Marlon, could upload them to our GitHub repository.

I downloaded the dataset from Kaggle.com and ran the full notebook in Jupyter with no errors. I checked our notebook against the professor's guidelines and found that normalization was missing. I added a log transform on glucose, using the training data only, with before-and-after histograms. For now it is a demonstration cell, and I asked the team whether to add it to the pipeline. I also confirmed that we used 5-fold cross-validation and that the data was split 70/30 in Python, not by hand.

I wrote the one-page proposal, the dataset source document, and my reflection journal. I also exported the notebook to PDF and sent the team a checklist of files for submission.

Doing this work taught me how important it is to check every step against the guidelines before submitting.

Student: Victor Djuma
Course: ITAI 1371

For this group midterm project, my individual contribution focused on working with the stroke prediction dataset and helping prepare it for machine learning. I worked with the project notebook in Jupyter and reviewed the different steps used to inspect, clean, transform, and prepare the dataset.

I worked with the original dataset containing 43,400 records and 12 columns. I helped verify the structure of the dataset, including the data types, missing values, duplicate records, and the distribution of the stroke target. The inspection showed that BMI and smoking status contained missing values and that the stroke target was highly imbalanced, with far fewer stroke cases than non-stroke cases.

I also worked with the 70/30 train-test split. The data was divided into 70% training data and 30% testing data using stratification. This kept approximately the same percentage of stroke cases in both sets. I learned that this split should happen before the main exploratory analysis so that decisions about data preparation are based on the training data rather than information from the testing data.

Another part of my contribution was reviewing the exploratory data analysis performed on the training data. I examined the class imbalance, missing values, numerical distributions, and relationships between some health-related features and the stroke target. This analysis helped guide the decisions used later during preprocessing.

I also worked through the preprocessing steps. Missing BMI values were handled using a median-based method, while missing smoking information was preserved in a way that did

not require guessing a patient's smoking history. Categorical features were converted into numerical form using one-hot encoding. Numerical features such as age, BMI, and average glucose level were scaled so they could be prepared for machine learning. I also reviewed the before-and-after results to make sure the preprocessing was working correctly.

I contributed to the class-imbalance section as well. The project calculates balanced class weights from the training labels so that stroke cases can receive more importance when models are trained later. This section helped demonstrate why a dataset with a very small number of positive cases needs special attention.

Another responsibility was making sure that the original dataset remained unchanged. The raw training and testing partitions were saved separately, and the processed datasets and preprocessing pipeline were prepared for use in the final project. This allows the project to keep the original data while also creating model-ready versions.

My main contribution was following and understanding the data-preparation process, checking the outputs, identifying problems such as missing values and class imbalance, and helping make sure that the dataset was ready for the next stage of the machine learning project. This work helped me improve my understanding of how Python and Jupyter can be used to turn raw data into data that is ready for machine learning.